

Exemplo de B-Spline com Forward Differences (em 2D)

Pontos de controle:

$$P1=(1,1)$$

$$P2=(2,2)$$

$$P3=(3,2)$$

$$P3=(4,1)$$

$$G_x = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$$

$$G_y = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 1 \end{pmatrix}$$

$$M_{BS} = \frac{1}{6} \begin{pmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 0 & 3 & 0 \\ 1 & 4 & 1 & 0 \end{pmatrix} = \begin{pmatrix} -1/6 & 1/2 & -1/2 & 1/6 \\ 1/2 & -1 & 1/2 & 0 \\ -1/2 & 0 & 1/2 & 0 \\ 1/6 & 2/3 & 1/6 & 0 \end{pmatrix}$$

$$E(\delta)=\begin{pmatrix} 0 & 0 & 0 & 1 \\ \delta^3 & \delta^2 & \delta & 0 \\ 6\delta^3 & 2\delta^2 & 0 & 0 \\ 6\delta^3 & 0 & 0 & 0 \end{pmatrix}$$

Vamos usar $\delta=0.1$

$$E(0.1)=\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0.001 & 0.01 & 0.1 & 0 \\ 0.006 & 0.02 & 0 & 0 \\ 0.006 & 0 & 0 & 0 \end{pmatrix}$$

*Para chamar a rotina do forward differences (2D),
precisamos calcular :*

$$\begin{pmatrix} f_x \\ \Delta f_x \\ \Delta^2 f_x \\ \Delta^3 f_x \end{pmatrix} \quad \begin{pmatrix} f_y \\ \Delta f_y \\ \Delta^2 f_y \\ \Delta^3 f_y \end{pmatrix}$$

Segundo a equação 5.39 (slides aula):

$$E(\delta) \cdot C = \begin{pmatrix} f \\ \Delta f \\ \Delta^2 f \\ \Delta^3 f \end{pmatrix}$$

*A matriz C é calculada para
cada coordenada com:*

$$\begin{aligned} C_x &= M_{BS} \cdot G_x \\ C_y &= M_{BS} \cdot G_y \\ &\text{(ver slide 1)} \end{aligned}$$

$$C_x = M_{BS} \cdot G_x = \begin{pmatrix} -1/6 + 1 - 3/2 + 2/3 \\ 1/2 - 2 + 3/2 \\ -1/2 + 3/2 \\ 1/6 + 4/3 + 1/2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix}$$

$$C_y = M_{BS} \cdot G_y = \begin{pmatrix} -1/6 + 1 - 1 + 1/6 \\ 1/2 - 2 + 1 \\ -1/2 + 1 \\ 1/6 + 4/3 + 1/3 \end{pmatrix} = \begin{pmatrix} 0 \\ -1/2 \\ 1/2 \\ 1.833 \end{pmatrix}$$

Então:

$$\begin{pmatrix} f_x \\ \Delta f_x \\ \Delta^2 f_x \\ \Delta^3 f_x \end{pmatrix} = E(0.1) \cdot C_x = \begin{pmatrix} 2 \\ 0.1 \\ 0 \\ 0 \end{pmatrix} \quad e \quad \begin{pmatrix} f_y \\ \Delta f_y \\ \Delta^2 f_y \\ \Delta^3 f_y \end{pmatrix} = E(0.1) \cdot C_y = \begin{pmatrix} 1.833 \\ 0.045 \\ -0.01 \\ 0 \end{pmatrix}$$

Rotina: fwdDiff(n, x, dx, d2x, d3x, y, dy, d2y, d3y);
(ver pseudocódigo nos slides)

Invocação para exemplo em questão:

$$n = 1/\delta = 10$$

$$x = 2; dx = 0.1; d2x = 0; d3x = 0;$$

$$y = 1.833; dy = 0.045; d2y = -0.01; d3y = 0;$$

fwdDiff(10, 2, 0.1, 0, 0, 1.833, 0.045, -0.01, 0);

fwdDiff(10, 2, 0.1, 0, 0, 1.833, 0.045, -0.01, 0);

n	x	dx	d2x	d3x	y	dy	d2y	d3y	Segmento de reta desenhado
0	2	0.1	0	0	1.833	0.045	-0.01	0	(estado inicial)
1	2.1	0.1	0	0	1.878	0.035	-0.01	0	(2, 1.833) → (2.1, 1.878)
2	2.2	0.1	0	0	1.923	0.025	-0.01	0	(2.1, 1.878) → (2.2, 1.923)
3	2.3	0.1	0	0	1.958	0.015	-0.01	0	(2.2, 1.923) → (2.3, 1.958)
4	2.4	0.1	0	0	1.983	0.005	-0.01	0	(2.3, 1.958) → (2.4, 1.983)
5	2.5	0.1	0	0	1.988	-0.005	-0.01	0	(2.4, 1.983) → (2.5, 1.988)
6	2.6	0.1	0	0	1.993	-0.015	-0.01	0	(2.5, 1.988) → (2.6, 1.993)
7	2.7	0.1	0	0	1.988	-0.025	-0.01	0	(2.6, 1.993) → (2.7, 1.988)
8	2.8	0.1	0	0	1.983	-0.035	-0.01	0	(2.7, 1.988) → (2.8, 1.983)
9	2.9	0.1	0	0	1.958	-0.045	-0.01	0	(2.8, 1.983) → (2.9, 1.958)
10	3.0	0.1	0	0	1.923	-0.055	-0.01	0	fim

(valores ao final de cada iteração)